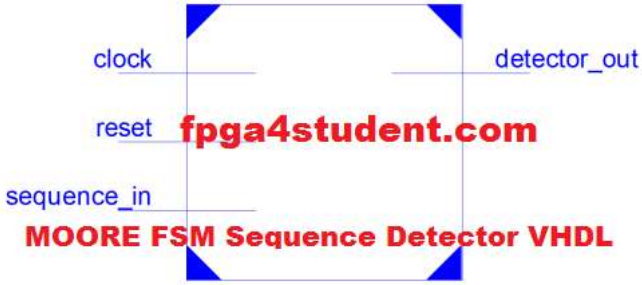
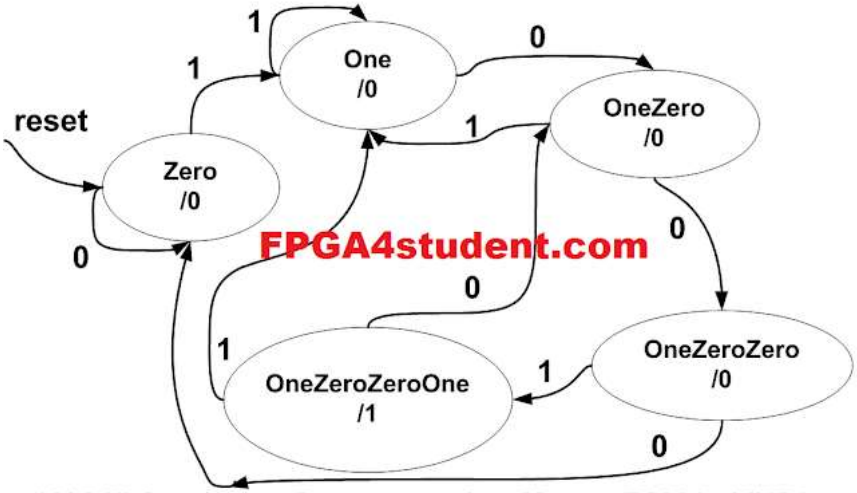
Full VHDL code for Moore FSM Sequence Detector

Last time, I presented a Verilog code together with Testbench for Sequence Detector using FSM. The sequence being detected was "1011".

This VHDL project presents a full VHDL code for Moore FSM Sequence Detector. A VHDL Testbench is also provided for simulation. The sequence to be detected is "1001".



The Moore FSM state diagram for the sequence detector is shown in the following figure.



VHDL code for Moore FSM Sequence Detector is designed based on Moore FSM's state diagram and block diagram:

```
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-- VHDL project: VHDL code for Sequence Detector using Moore FSM
-- The sequence being detected is "1001" or One Zero Zero One
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity VHDL_MOORE_FSM_Sequence_Detector is
port (
 clock: in std_logic; --- clock signal
 reset: in std_logic; -- reset input
 sequence_in: in std_logic; -- binary sequence input
 detector_out: out std_logic -- output of the VHDL sequence detector
);
end VHDL_MOORE_FSM_Sequence_Detector;

architecture Behavioral of VHDL_MOORE_FSM_Sequence_Detector is
type MOORE_FSM is (Zero, One, OneZero, OneZeroZero, OneZeroZeroOne);
signal current_state, next_state: MOORE_FSM;

begin
-- Sequential memory of the VHDL MOORE FSM Sequence Detector
```

```
process(clock,reset)
begin
 if(reset='1') then
  current_state <= Zero;
 elsif(rising_edge(clock)) then
  current_state <= next_state;
 end if;
end process;
-- Next state logic of the VHDL MOORE FSM Sequence Detector
-- Combinational logic
process(current_state,sequence_in)
begin
 case(current_state) is
 when Zero =>
  if(sequence_in='1') then
   -- "1"
   next_state <= One;
  else
   next_state <= Zero;
  end if;
 when One =>
  if(sequence_in='0') then
   -- "10"
   next_state <= OneZero;
  else
   next_state <= One;
  end if;
 when OneZero =>
  if(sequence_in='0') then
   -- "100"
   next_state <= OneZeroZero;
  else
   next_state <= One;
  end if;
 when OneZeroZero =>
  if(sequence_in='1') then
   -- "1001"
   next_state <= OneZeroZeroOne;
  else
   next_state <= Zero;
  end if;
 when OneZeroZeroOne =>
  if(sequence_in='1') then
   next_state <= One;
  else
   next_state <= OneZero;
  end if;
 end case;
end process;
-- Output logic of the VHDL MOORE FSM Sequence Detector
process(current_state)
begin
 case current_state is
 when Zero =>
  detector_out <= '0';
 when One =>
  detector_out <= '0';
 when OneZero =>
  detector_out <= '0';
 when OneZeroZero =>
  detector_out <= '0';
 when OneZeroZeroOne =>
  detector_out <= '1';
```

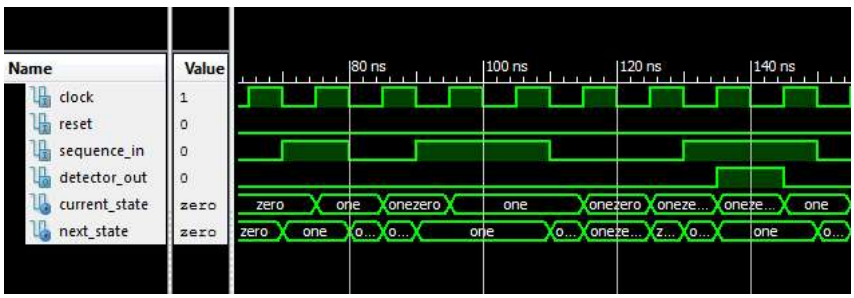
```
 end case;
end process;
end Behavioral;
```

VHDL Testbench for Sequence Detector using Moore FSM:

```
-- fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL code for Sequence Detector using Moore FSM
-- VHDL testbench for Moore FSM Sequence Detector
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY tb_VHDL_Moore_FSM_Sequence_Detector IS
END tb_VHDL_Moore_FSM_Sequence_Detector;

ARCHITECTURE behavior OF tb_VHDL_Moore_FSM_Sequence_Detector IS

    -- Component Declaration for the Moore FSM Sequence Detector in VHDL

    COMPONENT VHDL_MOORE_FSM_Sequence_Detector
    PORT(
         clock : IN  std_logic;
         reset : IN  std_logic;
         sequence_in : IN  std_logic;
         detector_out : OUT  std_logic
        );
    END COMPONENT;


   --Inputs
   signal clock : std_logic := '0';
   signal reset : std_logic := '0';
   signal sequence_in : std_logic := '0';

  --Outputs
   signal detector_out : std_logic;

   -- Clock period definitions
   constant clock_period : time := 10 ns;

BEGIN

 -- Instantiate the Moore FSM Sequence Detector in VHDL
   uut: VHDL_MOORE_FSM_Sequence_Detector PORT MAP (
          clock => clock,
          reset => reset,
          sequence_in => sequence_in,
          detector_out => detector_out
        );

   -- Clock process definitions
   clock_process :process
   begin
  clock <= '0';
  wait for clock_period/2;
  clock <= '1';
  wait for clock_period/2;
   end process;


   -- Stimulus process
   stim_proc: process
   begin
      -- hold reset state for 100 ns.
```

```
    sequence_in <= '0';
    reset <= '1';
    -- Wait 100 ns for global reset to finish
    wait for 30 ns;
        reset <= '0';
    wait for 40 ns;
    sequence_in <= '1';
    wait for 10 ns;
    sequence_in <= '0';
    wait for 10 ns;
    sequence_in <= '1';
    wait for 20 ns;
    sequence_in <= '0';
    wait for 20 ns;
    sequence_in <= '1';
    wait for 20 ns;
    sequence_in <= '0';
        -- insert stimulus here
        wait;
     end process;

END;
```

Simulation Waveform for Moore FSM Sequence Detector in VHDL:



As shown in the simulation waveform of the VHDL Moore FSM sequence detector, the detector output only goes high when the sequence "1001" is detected.

Verilog code for Moore FSM Sequence Detector: here.

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